



Add Voice Cloning in TTS (120 minutes)

The problem

Kokoro-82M is a frozen, CPU-real-time TTS model. Every voice it speaks in is nothing but a small **style tensor** handed to the model at synthesis time. The 54 stock voices are just 54 saved tensors.

Create a ~1 minute recording of your own voice. **Find the style tensor that makes Kokoro sound like you.** No training. No gradients required. The model stays frozen; you search the voice space.

Your files (tts_handout.zip)

None

<code>kokoro_assets/</code>	model weights + config + all 54 stock voice
<code>tensors (offline)</code>	
<code>reference/</code>	8 wav clips of the target speaker + their
<code>transcripts</code>	
<code>starter/</code>	
<code>synth.py</code>	text + style tensor -> wav (offline Kokoro
<code>wrapper)</code>	
<code>similarity.py</code>	Resemblyzer speaker-similarity utilities
<code>blend.py</code>	baseline: rank stock voices vs target, naive
<code>2-voice blend</code>	
<code>search.py</code>	random-walk search skeleton (the TODOs are
<code>your hour)</code>	

Setup is two commands, published in advance: `pip install --no-index --find-links wheels/ kokoro resemblyzer soundfile` plus `espeak-ng` installed on your machine.

The task

Produce `voice.pt` - a tensor with the same shape as the stock voice tensors - such that Kokoro speaking WITH YOUR TENSOR sounds like the target speaker.

The obvious ladder:



1. Rank the 54 stock voices by similarity to the target. (blend.py does this - it is your baseline to beat.)
2. Blend voices: convex combinations of the best few. Search the weights.
3. Perturbation search: random walk / genetic / CMA-ES directly in tensor space, scored by a fitness function YOU design.
4. Stretch: gradient descent on the tensor through the frozen model with a spectral loss.

Two hints we will ask about later: (a) a fitness function that is ONLY speaker-similarity can be fooled - the search will happily drift into degraded, raspy audio that the embedding loves and your ears hate. Listen to your candidates. (b) The 256 style dimensions may not all do the same kind of work - investigate.

Scoring (hidden)

We take your `voice.pt` and synthesize **sentences you have never seen**, then measure:

1. **Speaker similarity** - cosine similarity between your synthesized audio and held-out recordings of the target (our speaker model, our clips). Higher is better. This is the headline number.
2. **Intelligibility gate** - we transcribe your audio with an ASR model; if word error rate exceeds 25%, your similarity score is disqualified. A voice that matches the speaker but mangles the words is worthless.

Your baseline: the best single stock voice, scored the same way. Beat it decisively.

Deliverables

1. `SUMMARY.html` - A coding agent created HTML file, that summarizes your work and how you went about the entire research, and also summarizes `runlog.md` and `nodes.md` and why your approach, you think, is the best.
2. `voice.pt`
3. Your search code, runnable
4. `RUNLOG.md` - per search run: fitness design, settings, score, what you heard when you listened, what you changed. **Graded.**
5. `NOTES.md` - max 10 sentences: your fitness function, your best result, and your explanation of where and WHY the similarity plateaued.

Rules

- CPU only. Kokoro stays frozen - no fine-tuning, no other TTS or voice models. Resemblyzer is the only pretrained helper allowed.
- Everything offline; the kit contains all weights.



- AI coding assistants allowed. We know their unassisted score; your grade lives above it and in the 5-minute discussion after.
- The target speaker consented to this exercise. Using this kit to clone anyone who hasn't is off the table, in this room and outside it.

Suggested hour

- **0–10:** run `blend.py`, note the baseline similarity, LISTEN to the top stock voices next to the reference.
- **10–40:** design fitness, run searches. Synthesis is real-time, so a fitness call on 2–3 short sentences costs ~10–20s - budget a few hundred evaluations, cache everything you can.
- **40–50:** best candidates, listen hard, pick.
- **50–60:** RUNLOG.md, NOTES.md.